Jaya Krishna
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Sr Data Engineer

PROFESSIONAL SUMMARY:

- ✓ 9+ years of Software Development Life Cycle (SDLC) experience in Software, System Analysis, Design, Development, Testing, Deployment, Maintenance, Enhancements, Re-Engineering, Migration, Troubleshooting and Support of multi-tiered web applications in high performing environments.
- ✓ Hands-on experience in Hadoop Framework and its ecosystem including HDFS Architecture, Map Reduce Programming, Hive, Sqoop, HBase, Zookeeper, Couchbase, Storm, Solr, Oozie, Spark, Scala, Flume, and Kafka.
- ✓ Experience in batch processing and writing programs using Apache Spark for handling real-time analytics and Streaming of data.
- ✓ Good understanding of Zookeeper and Kafka for monitoring and managing Hadoop jobs and used Cloudera CDH5, CDH6 and CDP7 for monitoring and managing Hadoop cluster.
- ✓ Experience in Analysis, Design, Development and Big Data in Scala, python, Spark, Hadoop and HDFS environment.
- ✓ Expertise in deploying Hadoop Architecture and various components such as HDFS, Job Tracker, Task Tracker, Name Node, Data Node, and Map Reduce/ Yarn concepts.
- ✓ Good working experience on Spark (Spark Core Component, Spark RDD, Data Frames, Spark SQL, Spark Streaming, Spark MLlib, Spark GraphX, SparkR) with Scala and Kafka.
- ✓ Hands on experience in using other Amazon Web Services like S3, VPC, EC2, Autoscaling, RedShift, DynamoDB, Route53, RDS, Glacier, EMR.
- ✓ Experience on moving raw data between different systems using Apache NIFI.
- ✓ Experience in Datacenter Migration, Azure Data Factory (ADF) V2. Managing Database, Azure Data Platform services (Azure Data Lake (ADLS), Data Factory (ADF), Data Lake Analytics, Stream Analytics, Azure SQL DW, and HDInsight/Databricks.
- ✓ Experience in developing and designing POCs using Scala and deployed on the Yarn cluster, compared the performance of Spark with Hive and SQL/ Teradata.
- ✓ Strong understanding of RDD operations in Apache Spark i.e., Transformations, Actions, Persistence (Caching), Accumulators, Broadcast Variables, and Optimizing Broadcasts.
- ✓ In-depth understanding of Apache spark job execution Components like DAG, lineage graph, Dag Scheduler, Task scheduler, Stages, and task.
- ✓ Experience in migrating Python Machine learning modules to scalable, high-performance, and fault-tolerant distributed systems like Apache Spark.
- ✓ Expertise in developing multiple Spark jobs in Scala for data cleaning, pre-processing and aggregating.
- ✓ Expertise in working with Hive data warehouse, creating tables, data distribution by implementing partitioning and bucketing, writing and optimizing the HiveQL queries.
- ✓ Excellent understanding and knowledge of job workflow scheduling and locking tools/ services like Oozie and Zookeeper.
- ✓ Knowledge of ETL methods for data extraction, transformation and loading in corporate-wide ETL Solutions and Data Warehouse tools for reporting and data analysis.
- ✓ Comprehensive knowledge in Debugging, Optimizing and Performance Tuning of DB2, Oracle and MYSQL databases.
- ✓ Good experience with Informatica, AWS Glue for designing ETL Jobs for Processing of data.
- ✓ Experience with Azure transformation projects and implement ETL and data movement solutions using Azure Data Factory (ADF), and SSIS.
- ✓ Good understanding and knowledge of NoSQL databases like MongoDB, HBase and Cassandra.
- ✓ Experience in Monitoring Hadoop clusters using tools like Nagios, CloudWatch, and Cloudera Manager.
- ✓ Experience with operating systems such as Windows, Linux, RedHat, and UNIX.
- ✓ Experienced and skilled Agile Developer with a strong record of excellent teamwork and successful coding.
- ✓ Strong Problem Solving and Analytical skills and abilities to make Balanced Independent decisions.
- ✓ Exceptional ability to quickly master new concepts and capable of working in-group as well as independently with excellent communication skills.

TECHNICAL SKILLS:

Operating Systems: Linux, UNIX, Windows.
Development Methodologies: Agile/ Scrum, Waterfall.
IDEs: Eclipse, Net Beans, IntelliJ
Big Data Platforms: Hortonworks, Cloudera CDH5, CDH6, CDP7
Programming Languages: Python, Scala, SQL, PySpark, Unix Shell script
Hadoop Components: HDFS, Sqoop, Hive, MapReduce, YARN, Impala, Hue, Zookeeper
Spark Modules: Spark Core Component, Spark SQL, Spark Streaming, Spark MLlib, Spark GraphX, SparkR
RDBMS/NOSQL Databases: Oracle, MYSQL, Microsoft SQL Server, HBase, MongoDB, Cassandra, DB2
Cloud Technologies: Amazon Web Services (AWS), Azure Data Factory, Databricks
DevOps Tools: Maven, Jenkins, Git, GitHub
Visualization and Reporting: Power BI, Tableau

PROFESSIONAL EXPERIENCE:

Role: Sr. Data Engineer
Client: Wells Fargo, Newark, NJ **April 2024 to Present**
Responsibilities:

- ✓ Involved in all phases of Software Development Life Cycle (SDLC) such as requirements gathering, modelling, analysis, design, development, and testing.
- ✓ Developed and deployed Spark applications using Pyspark and Spark-SQL for data extraction, transformation, and aggregation from multiple formats for analyzing & transforming the data to uncover insights into the customer usage patterns
- ✓ Developed HIVE UDFs to incorporate external business logic into Hive script and developed join data set scripts using HIVE join operations.
- ✓ Created various Hive external tables, staging tables and joined the tables as per the requirement. Implemented static Partitioning, Dynamic partitioning and Bucketing.
- ✓ Worked with the Spark for optimization of the existing algorithms in Hadoop using Spark-SQL, PySpark, Pair RDD's, and YARN.
- ✓ Involved in migrating the client Datawarehouse architecture from on-premises into MS Azure cloud.
- ✓ Used Spark for interactive queries, processing of streaming data and integration with popular NoSQL database for huge volume of data.
- ✓ Worked with NoSQL databases like HBase in creating HBase tables to load large sets of semi-structured data coming from various sources.
- ✓ Created pipelines in ADF using linked services to extract, transform and load data from multiple sources like Azure SQL, Blob storage and Azure SQL Data warehouse.
- ✓ Involved in implementing Cluster for NoSQL tool HBase as a part of POC to address HBase limitations.
- ✓ Used Spark Data Frames Operations to perform required Validations in the data and to perform analytics on the Hive data.
- ✓ Developed Apache Spark applications for data processing from various streaming sources like kafka.
- ✓ Design and implement end-to-end data solutions (storage, integration, processing, and visualization) in Azure.
- ✓ Worked on architecture and components of Spark, and efficient in working with Spark Core, Spark SQL.
- ✓ Developed custom Kafka producer and consumer for different publishing and subscribing to Kafka topics.
- ✓ Migrated Map reduce jobs to Spark jobs to achieve better performance.
- ✓ Developed Python, Shell/ Perl Scripts and Power shell for automation purpose and Component unit testing using Azure Emulator.
- ✓ Used HUE for running Hive queries. Created partitions according to day using Hive to improve performance.
- ✓ Implement ad-hoc analysis solutions using Azure Data Lake Analytics/ Store, and HDInsight. #
- ✓ Developed Oozie workflow engine to run multiple Hive, Sqoop and Spark jobs.
- ✓ Implement ETL and data movement solutions using Azure Data Factory (ADF), and SSIS.
- ✓ Worked on auto scaling the instances to design cost effective, fault tolerant and highly reliable systems.

Environment: MS Azure, Azure SQL, Blob Storage, Azure Data Factory (ADF), HDInsight, Python, OLAP, Hadoop (HDFS, MapReduce), YARN, Spark, Spark-SQL, PySpark, Pair RDD's, Spark DataFrames, Hive, HBase, Oozie, Hue, Sqoop, Flume, Oracle, NIFI, Kafka.

Role: Data Engineer
Walmart, Bentonville, AR **Oct 2021 to March 2024**
Responsibilities:

- ✓ Analyzed stories, participated in grooming sessions and point estimation for development according to agile methodology.
- ✓ Developed Spark jobs using Scala and Python on top of Yarn/ MRv2 for interactive and Batch Analysis.
- ✓ Worked with the Spark for improving performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, PySpark, Pair RDD's, Spark YARN, and Spark MLLib.
- ✓ Involved in implementation of data movements from on-premises to cloud in MS Azure.
- ✓ Evaluated the performance of Apache Spark in analyzing genomic data.
- ✓ Developed a NIFI Workflow to pick up the data from SFTP server and send that to Kafka broker.
- ✓ Used Python - NumPy, SciPy, Pandas, NLTK, Matplotlib, Beautiful Soup, and Text Blob to finish the ETL process of clinical data for future NLP analysis.
- ✓ Migrated data from traditional database systems to Azure SQL databases.
- ✓ Used different Machine Learning modules for ruining the spark jobs on daily and weekly basis.
- ✓ Migrated complex Map reduce programs into Spark RDD transformations, actions.
- ✓ Designed and developed Oozie workflows to orchestrate Hive scripts, Sqoop.
- ✓ Involved in developing the Spark Framework to provide structure to data on the fly and process the data using Spark core API's, Data Frame, Spark-SQL and Scala Evaluated and improved application performance with Spark.
- ✓ Recreated existing application logic and functionality in the Azure Data Lake, Data Factory, and Azure SQL Database.
- ✓ Developed Spark scripts to import large files from Azure and imported the data from different sources like HDFS/HBase into Spark RDD.
- ✓ Extensively worked on Python and build the custom ingest framework.
- ✓ Design, build and deliver the operational and management tools, frameworks and processes for the Azure Data Lake and drive the implementations into the Azure Data Lake Cloud Operations team.
- ✓ Involved in working with Spark on top of YARN for interactive and batch analysis.
- ✓ Implemented Spark SQL to update queries based on the business requirements.
- ✓ Used Spark API over Hortonworks Hadoop YARN to perform analytics on data in HDFS.
- ✓ Designed Columns families in Cassandra and Ingested data from RDBMS, performed transformations and exported the data to Cassandra.
- ✓ Successfully migrated Data Pipeline jobs from Oozie to Airflow.
- ✓ Developed Scala scripts, UDFs using both Data frames/ SQL/ Data sets and RDD in Spark for Data Aggregation, queries and writing data back into OLTP system through Sqoop.
- ✓ Created Azure Stream Analytics Jobs to replication the real time data to load to Azure SQL Data warehouse.
- ✓ Implemented the real time streaming ingestion using Kafka and Spark Streaming.
- ✓ Developed visualizations and dashboards using Power BI.
- ✓ Created dashboards for analyzing POS data using Power BI.

Environment: Big Data, Spark, Python, NumPy, SciPy, Pandas, NLTK, Matplotlib, Beautiful Soup, Text Blob, MS Azure, Azure SQL, Azure Data Lake, Azure Data Factory, Scala, YARN, Spark Context, Spark-SQL, PySpark, Pair RDD's, Spark YARN, Spark MLLib, NiFi, Kafka, Oozie, OLAP, Power BI.

Azure Big Data Engineer
United Health Group, Eden Prairie, MA **April 2019 to Sept 2021**
Responsibilities:

- ✓ Used Azure Data Factory extensively for ingesting data from disparate source systems.
- ✓ Used Azure Data Factory as an orchestration tool for integrating data from upstream to downstream systems.
- ✓ Automated jobs using different triggers (Event, Scheduled and Tumbling) in ADF.
- ✓ Used Cosmos DB for storing catalog data and for event sourcing in order processing pipelines.
- ✓ Analyzed the data flow from different sources to target to provide the corresponding design Architecture in Azure environment.
- ✓ Created High level technical design documents and Application design document as per the requirements and delivered clear, well-communicated and complete design documents.
- ✓ Created DA specs and Mapping Data flow and provided the details to developer along with HLDs.

- ✓ Created Build definition and Release definition for Continuous Integration and Continuous Deployment.
- ✓ Created Application Interface Document for the downstream to create new interface to transfer and receive the files through Azure Data Share.
- ✓ Creating pipelines, data flows and complex data transformations and manipulations using ADF and PySpark with Data bricks
- ✓ Ingested data in mini-batches and performs RDD transformations on those mini-batches of data by using Spark Streaming to perform streaming analytics in Data bricks.
- ✓ Created, provisioned different Data bricks clusters needed for batch and continuous streaming data processing and installed the required libraries for the clusters. Integrated Azure Active Directory authentication to every Cosmos DB request sent and demoed feature to Stakeholders
- ✓ Created numerous pipelines in Azure using Azure Data Factory v2 to get the data from disparate source systems by using different Azure Activities like Move Transform, Copy, filter, for each, Data bricks etc.
- ✓ Created several Data bricks Spark Jobs with PySpark to perform several tables to table operations.
- ✓ Extensively used SQL Server Import and Export Data tool.
- ✓ Working with complex SQL, Stored Procedures, Triggers, and packages in large databases from various servers.

Environment: Hadoop, Hive, Oozie, Spark, Spark SQL, Python, PySpark, Azure Data Factory, Azure SQL, Azure Data bricks, Azure DW, BLOB storage, Scala, AWS, Linux, Maven, Oracle 11g/10g, Zookeeper, MySQL, Snowflake.

Role: Data Engineer
EPAM, Hyderabad, India **June 2016 to Nov 2018**
Responsibilities:

- ✓ Administered, maintained, provisioned, patched, and maintained Cloudera Hadoop clusters on Linux.
- ✓ Worked on analyzing Hadoop stack and different big data analytic tools including Hive, HBase database, and Sqoop.
- ✓ Utilized Python matplotlib and SciKit-Learn modules to generate the basic prototype visualization restored using visualization tools such as Tableau.
- ✓ Written multiple MapReduce programs to extract data for extraction, transformation, and aggregation from more than 20 sources having multiple file-formats including XML, JSON, CSV & other compressed file formats.
- ✓ Worked in AWS environment for development and deployment of Custom Hadoop Applications.
- ✓ Created Oozie workflows for Hadoop-based jobs including Sqoop, Hive.
- ✓ Involved in file movements between HDFS and AWS S3.
- ✓ Created Hive External tables and loaded the data into tables and query data using HQL.
- ✓ Performed data validation on the data ingested using MapReduce by building a custom model to filter all the invalid data and cleanse the data.
- ✓ Handled the importing of data from various data sources, performed transformations using Hive, and Map-Reduce, loaded data into HDFS, and extracted data from MySQL into HDFS using Sqoop.
- ✓ Transferred the data using the Informatica tool from AWS S3 to AWS Redshift.
- ✓ Wrote HiveQL queries by configuring a number of reducers and mappers in the query needed for the output.
- ✓ Transferred data from Hive using HCatalog, transferred relational database using Sqoop.
- ✓ Involved in Extract Transfer and Load (ETL) process using SSIS and generated reports using SSRS.
- ✓ Analyzed the data by performing Hive queries (HiveQL)..
- ✓ Cluster coordination services through Zookeeper. Installed and configured Hive and written Hive UDFs.

Environment: Hadoop, Hive, HBase, Sqoop, Python, Matplotlib, Scikit Learn, Tableau, HDFS, Map Reduce, AWS, S3, RedShift, MySQL, OLAP, Oozie, Sqoop, HQL, SSIS, SSRS, MS SQL Server, HiveQL.